Due: Thursday April 21st, latest at 2PM

– – – – – – – – – – – – – – – ⋆ – – – – – – – – – – – – –

- This problem set is on electromagnetic induction and on using Maxwell's equations.

– – – – – – – – – – – – – – – ⋆ – – – – – – – – – – – – –

1. A square loop with sides of length L lies in the x-y plane in a region in which the magnetic field points in the z-direction and changes over time as

$$\mathbf{B}(t) = B_0 e^{-5t/t_0} \hat{k}$$

   (a) [**5 pts**] Find the magnitude of the EMF induced in the wire.

   (b) [**3 pts**] You are looking from 'above' onto the x-y plane, so that the positive z-axis points out of the paper toward you. What is the direction of the induced current in the loop? (Counterclockwise or clockwise?) You will only get credits for your answer if you justify it using Lenz's law.

2. An electromagnetic system is described by the time-dependent fields

$$\mathbf{E} = -Cy\cos(\omega t)\hat{k}\ ; \quad \mathbf{B} = B_0 \sin(\omega t)\hat{\imath}$$

   where B_0 and ω are positive constants.

   (a) [**7 pts**] Using Maxwell's third equation (which concerns the curl of the electric field), express the constant C as a function of B_0 and ω.

   (b) [**6 pts**] Find the current density $\mathbf{J}$. Your answer should contain B_0 and ω, not C.

3. [**8 pts**] An electromagnetic wave has an electric field given by

$$\mathbf{E} = E_0 \cos(2\pi f t)\sin\left(\frac{2\pi z}{\lambda}\right)\hat{\imath}$$

   Use Maxwell's equations to calculate the associated magnetic field $\mathbf{B}$. (You will need to figure out which one of them, or which ones, to use.)

4. **[7 pts]**

   The magnetic field in some region is given by

$$\mathbf{B} = \frac{\mu_0}{\pi}\left(\frac{Cy}{x^2+y^2}\hat{i} + \frac{2x}{x^2+y^2}\hat{j}\right)$$

   Using Maxwell's second equation (which concerns the divergence of the magnetic field), determine the constant C.

5. A long solenoid has n turns per unit length and radius R. The current through the solenoid increases with time: $I(t) = \alpha t$. We will use Maxwell's 3rd equation (Faraday's law) in integral form, to calculate the magnitude of the electric field created by electromagnetic induction. The calculation is silmilar to using Ampere's law to calculate the magnetic field due to a thick wire.

   (a) **[7 pts]** Calculate the electric field at a distance $r > R$ from the axis of the solenoid (outside the solenoid).

   (b) **[7 pts]** Calculate the electric field at a distance $r < R$ from the axis (inside the solenoid).